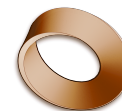


Name: _____

Date: _____

Mr. Rawson • Y9 MYP Physics



1.2 – Modelling a Spring

Unit 1: Oscillations & Waves

The question: *can we predict how far a spring stretches before we pull it?* Last lesson you drew a model of a pendulum. Today you build a model from measurements: measure, tabulate, plot, find the relationship. Nobody is going to tell you the answer — your data is the answer.

Part 1 — The system

1) **Draw** the model of your spring hanging from the clamp with a mass on the end. Label the fixed point, and draw the force arrows — remember that gravity is a vector that does not change on Earth.

Part 2 — Planning the investigation

Variables: The **independent** variable is the one you change on purpose. The **dependent** variable is the one you measure. **Control** variables are kept the same so the test stays fair.

2) **State** the independent variable — what will you change, one step at a time?

3) **State** the dependent variable — what will you measure each time?

4) **State** two things you must keep the same for the test to be fair.

Name: _____

Date: _____

Part 3 — Collecting the data

Method

1. Hang your spring from the clamp. Measure its **natural length** — its length with nothing on it — and write it here: natural length = _____ cm
2. Add the hanger and masses **one at a time**. Each 100 g of mass pulls on the spring with a force of about **1 N**.
3. After each mass, measure the **total length** and work out the extension straight away:

$$\text{extension} = \text{total length} - \text{natural length}$$
4. Take at least **six** readings. **Stop adding masses** if the coils of the spring stay separated when you look at it — your teacher will tell you the maximum load for your spring.

Safety: eye protection on; keep feet out from under the masses.

Mass added / g	Force / N	Total length / cm	Extension / cm
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0

The first row is your check: with no mass added, the extension must be 0.

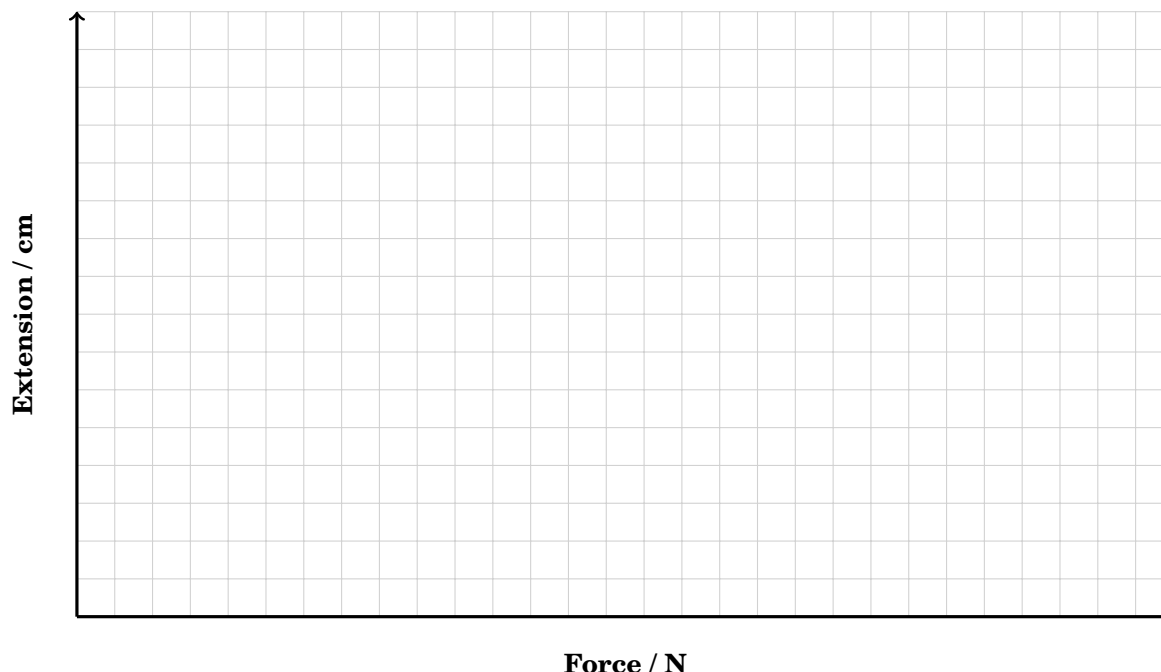
Finished early? Swap for a different spring, measure its natural length, and fill a second table:

Mass added / g	Force / N	Total length / cm	Extension / cm
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0

Part 4 — The graph

5) **Plot** your results: force on the horizontal axis, extension on the vertical axis. Choose scales that use most of the grid, label both axes with the quantity and its unit, then draw a single straight **line of best fit** with a ruler — do not join the dots.



Part 5 — The same model, built by a machine

Spreadsheet

1. Open a new spreadsheet. Type two column headings: Force (N) and Extension (cm). Type your readings underneath.
2. Select both columns and **insert a chart**. Choose a **scatter** chart — points only, no joining lines.
3. Add a **trendline** (the spreadsheet's name for a line of best fit). Compare it with the line you drew by eye.

Part 6 — Conclusion

Describe: Give a detailed account or picture of a situation, event, pattern or process.

6) **Describe** the pattern your graph shows. What happens to the extension when the force doubles?

7) **Predict** the extension for a force you did *not* test, using your line of best fit. Show on the graph how you read it off.

8) **Compare** your graph with a pair who tested a different spring. What is the same about the two graphs, and what is different?
